I. Overview
    A. Goal: Compare MD sims to Seitz model
    B. Gaps in past work
        1. assumptions made in bridge from model fluids to reality
            a) e.g. — MD sims of LJ fluids compared to Seitz model for noble liquids
        2. details of sims not always clear
            a) NPE sims often used, but barostat likely impacts bubble growth
            b) time steps, cutoffs, etc, often not specified
        3. key test cases missing
    C. Fill the gaps by
        1. compare MD sims of LJ fluids to Seitz model built on thermophysical parameters of LJ fluids *measured in the same MD sim framework*
        2. Validate NPE vs NVE sims (do both, verify that box is big enough that it doesn't matter, or at least bracket result)
        3. Look at intermediate cases
            a) Cavitation versus hotspike – do get critical radius correct?
            b) Small spherical energy depositions – is there fundamental correction to Seitz needed before tracks are even considered?

II. Simulated Thermophysical Properties of LJ fluids
    A. Need surface tension, vapor pressure, and heat of vaporization to calculate Seitz Threshold
    B. MD sim technique to measure surface tension, vapor pressure, heat of vaporization, and gas/liquid densities, all versus temperature
    C. Results
        1. versus temperature for one LJ fluid
        2. comparison between different LJ fluids (i.e. different cutoff styles and distances)
    D. Comparison to Argon
        1. anchoring units Tcrit, Pcrit
        2. anchoring units using rho_liquid / rho_gas
        3. Choose most argon-like model for rest of studies

III. Preparing superheated state
    A. Homogeneous state, slightly below liquid density (pressure below vapor pressure)
        1. Prepare in NVT mode
        2. How long is it stable for?
        3. Can we measure homogeneous nucleation? (at some densities, should be possible)

IV. Simulated cavitation – measure critical radius via simulation
    A. Remove atoms from superheated state, creating "vapor" bubble of some radius
    B. Evolve until either:
        1. return to uniform state (liquid)
        2. phase separation

3. Determine: How long does simulation have to run for to distinguish (1) from (2)?

C. Repeat for NVT, NVE, NPE, NPT, different box sizes
    1. want to know: how big does box have to be for NVE to work well?
        a) meaning: liquid has similar T, P to start if bubble collapses, and bubble has room to grow if it doesn't

V. Simulated hot spikes
    A. Simplest spike – up to 100x temp in spherical region, evolve in NVE mode
        1. box size chosen based on work in section IV
        2. time step shrinks by x10, at least for early evolution, since atoms are faster
    B. Vary:
        1. Superheated state
            a) kT
            b) Density
        2. Hot spike
            a) temp
            b) radius
    C. Discuss
        1. How does required energy deposition (temp * volume) depend on radius?
        2. Does required energy deposition match Seitz calculation?
        3. Compare to calibration data, other MD sims
            a) XeBC result
            b) UAlberta paper
            c) Zurich paper
            d) potential to verify w/ SBC-LAr10